Fourier Series Representation of Periodic Signals

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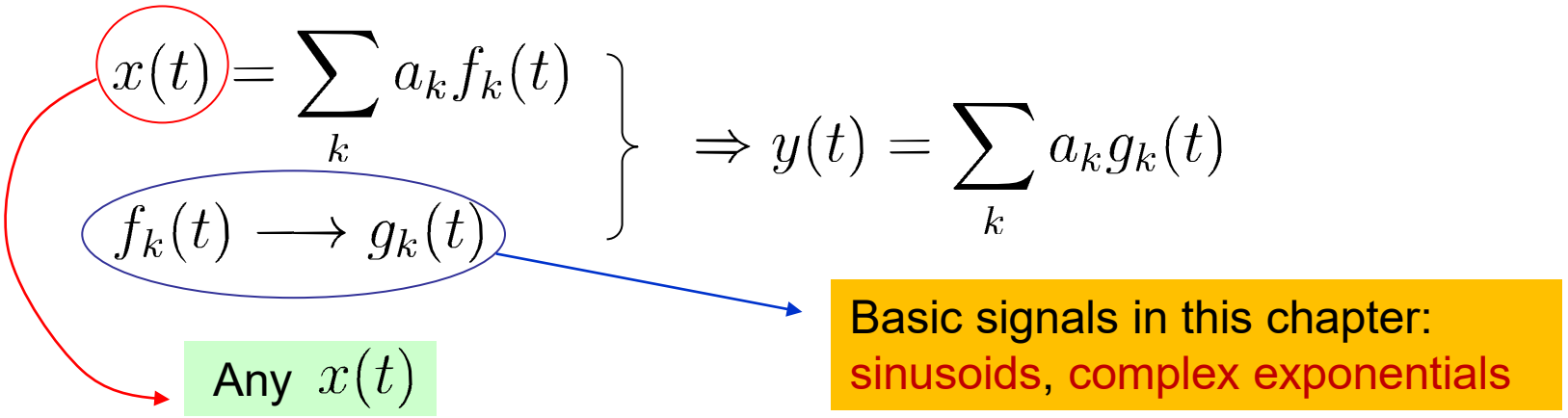
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- The headaches of time-domain analysis:
  - Some feature of the signal is not obvious (so we need to transform the signal to frequency domain)
  - The computation of the convolution is complicated (in frequency domain it is multiplication)
  - To get the explicit output of the system one needs to solve LCCDE (in frequency domain it is more easier, will see)

- Motivations:

  Because of the superposition property of LTI system, if we could represent ANY input as a linear combination of basic signals, then the output is a linear combination of the individual responses to each of the basic signals.



Jean Baptiste Joseph Fourier
(March 21, 1768 ~May 16, 1830)



- French mathematician and physicist who is best known for initiating the investigation of Fourier series and their application to problems of heat flow.
- The Fourier transform is also named in his honor
- In 1807, "any periodical signals can be represented by harmonically related sinusoids series"
- Lagrange opposed.
- In 1822, 15 years after Fourier had first presented his results, his work finally appeared as an essay.
- In 1829, Dirichelet presents the condition for the convergence of Fourier series
- In 1830, Fourier passed away

1960s, the invention of FFT

The Two Most Important Contributions of Fourier

- Periodical signals can be represented by the weighted sum of harmonically related complex exponentials

  ——— Fourier series

- Non-periodical signals can be represented by the weighed integration of (not harmonically related) complex exponentials ——— Fourier transform

Holds for both continuous-time and discrete-time signals

The Response of LTI Systems to Complex Exponentials

- Selection of basic signals
  - The basic signals should be simple and the corresponding system's output is also simple
  - The basic signals should be "distinguishable" either in time domain or frequency domain, i.e., being "slim"
  - The basic signals should be universally used to build any signals $x(t)$
- Complex exponentials e^{st} or z^n are selected as basic signals
  - It satisfies the above requirements
  - It even provides the same complex exponential with only a change in amplitude

$$\text{C-T:} \quad \mathrm{e}^{st} \longrightarrow H(s)\mathrm{e}^{st},$$

$$\text{D-T:} \quad z^n \longrightarrow H(z)z^n.$$

- Continuous time:

$$e^{st} \rightarrow \boxed{h(t)} \rightarrow y(t)$$

  - From definition

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}d\tau$$

$$= e^{st}\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau = e^{st}H(s)$$

eigenfunction eigenvalue

  - The response of LTI system to complex exponentials is very simple. In fact, with only an amplitude change $H(s)$ (constant)

- The complex exponentials e^{st} are the eigenfunction of any LTI systems, while $H(s)$ are the corresponding eigenvalues

- If we represent $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{s_k t}$

  then the response is $y(t) = \sum_{k=-\infty}^{\infty} a_k H(s_k) e^{s_k t}$

  The only thing we need to know is

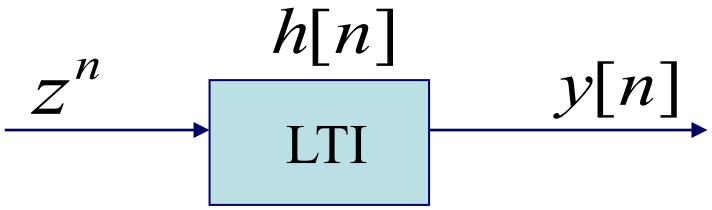
$$H(s) = \int_{-\infty}^{\infty} h(t) e^{-s\tau} d\tau$$

Compare with convolution

Fourier analysis only focuses on $s = j\omega$

The Response of LTI Systems to Complex Exponentials

- Discrete time:



eigenfunction

eigenvalue

  - From definition

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] z^{n-k} = z^n \sum_{k=-\infty}^{\infty} h[k] z^{-k} = z^n H(z)$$

  - The complex exponentials z^n are the eigenfunction of any LTI systems, while $H(z)$ are the corresponding eigenvalues
  - Similarly

$$x[n] = \sum_{k=-\infty}^{\infty} a_k z_k^n \quad \Rightarrow \quad y[n] = \sum_{k=-\infty}^{\infty} a_k H(z_k) z_k^n$$

  - Need to know $H(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-n}$

Fourier analysis only focuses on $z = e^{j\omega}$

- Eigenvalues decomposition
  - The eigenvalue equation in its matrix representation is

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$$

  - where $\mathbf{u}$ is an eigenvector, λ is eigenvalue.
- Equivalently

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{u} = \mathbf{0}$$

- For a non-zero eigenvector, it is equivalent to

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

  - which is a polynomial to be solved.

- Example: $y(t) = x(t-3)$ and $x(t) = \cos(4t) + \cos(7t)$
  - Regular way: $y(t) = \cos(4(t-3)) + \cos(7(t-3))$
  - New way:

- Example: $y(t) = x(t-3)$ and $x(t) = \cos(4t) + \cos(7t)$
  - Regular way: $y(t) = \cos(4(t-3)) + \cos(7(t-3))$
  - New way:

$$x(t) = \frac{1}{2}e^{j4t} + \frac{1}{2}e^{-j4t} + \frac{1}{2}e^{j7t} + \frac{1}{2}e^{-j7t}$$

$$h(t) = \delta(t-3) \qquad H(s) = \int_{-\infty}^{\infty} \delta(\tau - 3)e^{-s\tau}d\tau = e^{-3s}$$

$$H(4j) = e^{-12j},\ H(-4j) = e^{12j},\ H(7j) = e^{-21j},\ H(-7j) = e^{21j}$$

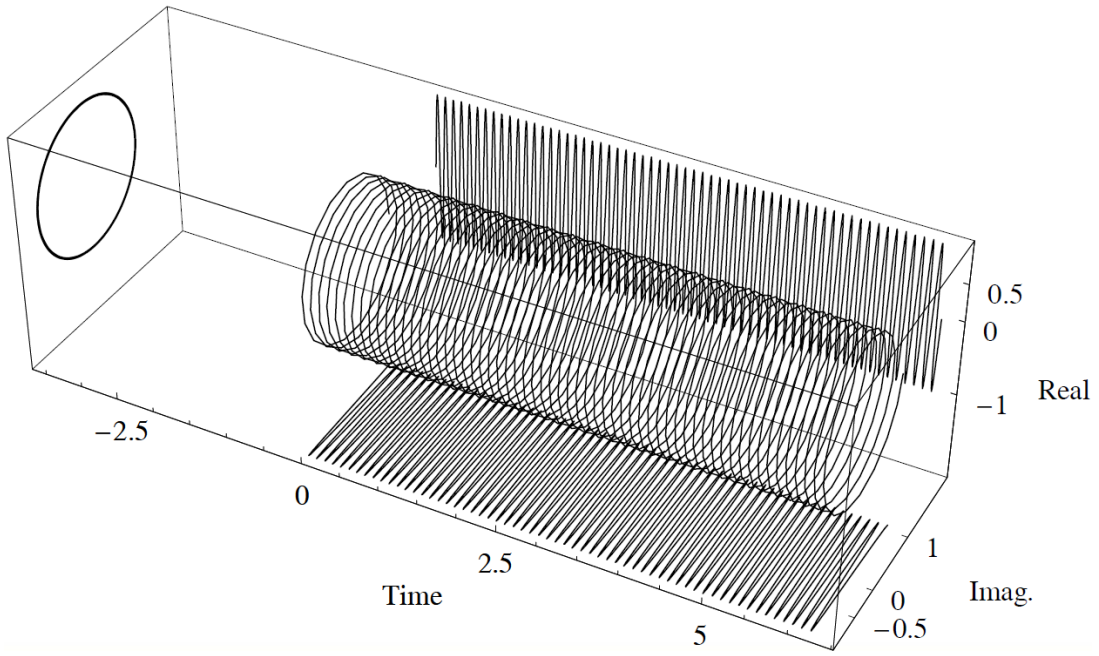
$$y(t) = \frac{1}{2}e^{-12j}e^{j4t} + \frac{1}{2}e^{12j}e^{-j4t} + \frac{1}{2}e^{-21j}e^{j7t} + \frac{1}{2}e^{21j}e^{-j7t}$$

$$= \cos(4(t-3)) + \cos(7(t-3))$$

What is the period of this signal?

Fourier Series Representation of Continuous-Time Periodic Signals

- Previous review:
  - Periodic signals $x(t) = x(t+T)$, with fundamental frequency $\omega_0 = 2\pi/T$
  - The basic periodic signal $e^{j\omega_0 t}$, and the harmonically related complex exponentials $\phi_k(t) = e^{jk\omega_0 t}, \ k = 0, \pm 1, \pm 2$



Fourier Series Representation of Continuous-Time Periodic Signals

- Let's see the follow signals

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \phi_k(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Manually made

FS series, if holds

  - The period is $T = 2\pi/\omega_0$
  - The kth harmonic components has frequency $k\omega_0$
  - The kth harmonic components has period T/k
  - a_k is Fourier series coefficient
- Clearly, Fourier series can represent the periodic signals
  - Reversely, could periodic signals be represented by the linear combination of harmonic components?

- Determine the Fourier series for a period signal
  - Multiply both sides with $e^{-jn\omega_0 t}$

Assume FS representation holds

$$x(t)e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t}$$

  - Integrate over the fundamental period $T = 2\pi/\omega_0$

$$\int_0^T x(t)e^{-jn\omega_0 t}dt = \int_0^T \sum_{k=-\infty}^{\infty} a_k e^{j(k-n)\omega_0 t}dt = Ta_n$$

Can be any interval of length T

$$\int_0^T e^{j(k-n)\omega_0 t}dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases}$$

$$\Rightarrow \quad a_n = \frac{1}{T}\int_0^T x(t)e^{-jn\omega_0 t}dt$$

Does this mean we've solved the issue??

- To summarize, if the periodic signal $x(t)$ has a Fourier series representation, then

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

Synthesis Equation

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_T x(t)e^{-jk(2\pi/T)t}dt$$

Analysis Equation

  - $a_0 = \frac{1}{T}\int_T x(t)dt$ is also called the DC component of $x(t)$

- For real signals $x(t) = x^*(t)$

$$\sum_{k=-\infty}^{\infty} a_k^* e^{-jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad \Longrightarrow \quad a_k = a_{-k}^*$$

  - Then

$$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t}] = a_0 + \sum_{k=1}^{\infty} 2\Re\{a_k e^{jk\omega_0 t}\}$$

Polar form $\quad a_k = A_k e^{j\theta_k} \qquad x(t) = a_0 + 2\sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k)$

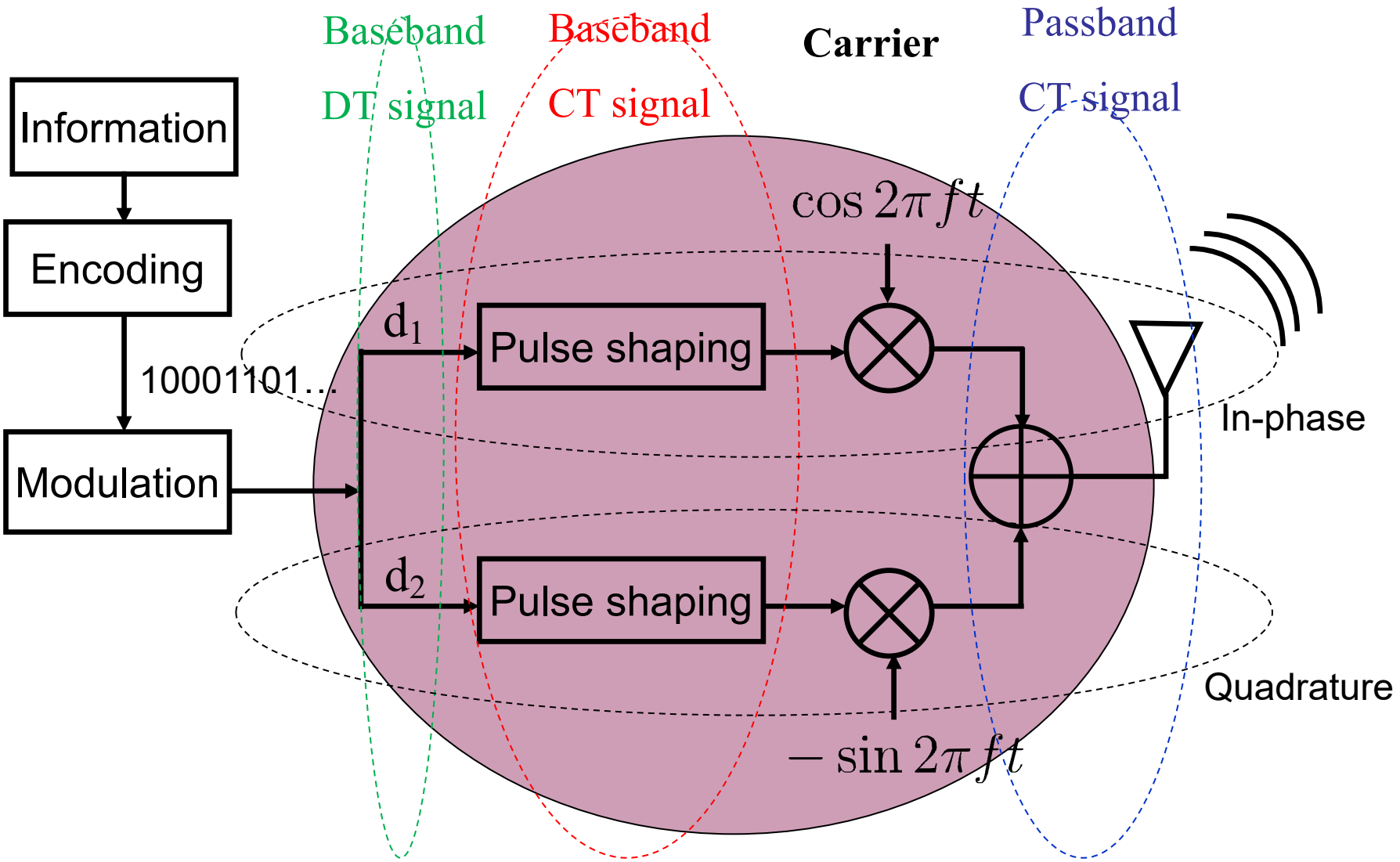
Rectangular form $\quad a_k = B_k + jC_k$

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} (B_k \cos k\omega_0 t - C_k \sin k\omega_0 t)$$



Example: Digital Communications

Transmitter of Digital Communications

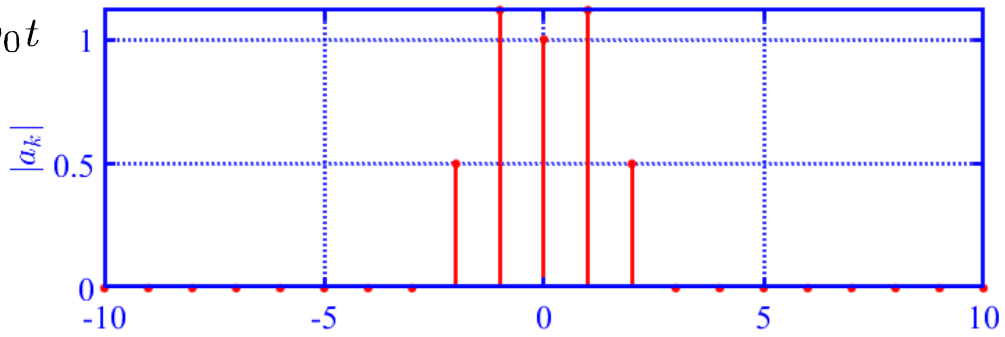


- Example: $x(t) = 1 + \sin \omega_0 t + 2 \cos \omega_0 t + \cos(2\omega_0 t + \pi/4)$

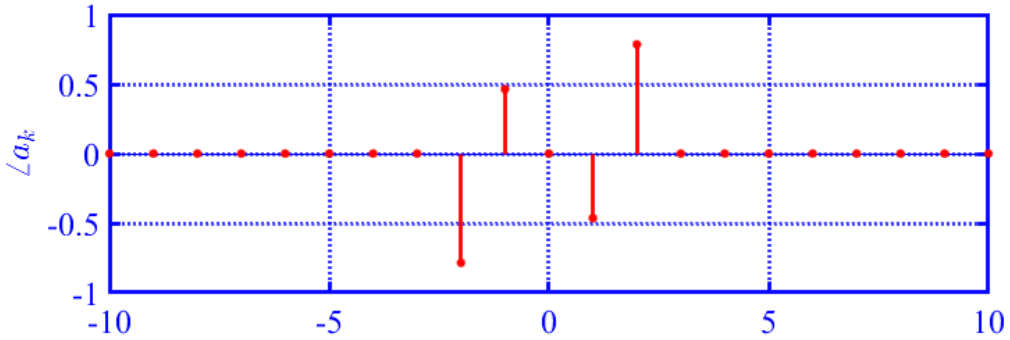
- Example: $x(t) = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos(2\omega_0 t + \pi/4)$

$$x(t) = 1 + \left(1 + \frac{1}{2j}\right) e^{j\omega_0 t} + \left(1 - \frac{1}{2j}\right) e^{-j\omega_0 t} + \left(\frac{1}{2} e^{j\pi/4}\right) e^{j2\omega_0 t} + \left(\frac{1}{2} e^{-j\pi/4}\right) e^{-j2\omega_0 t}$$

magnitude



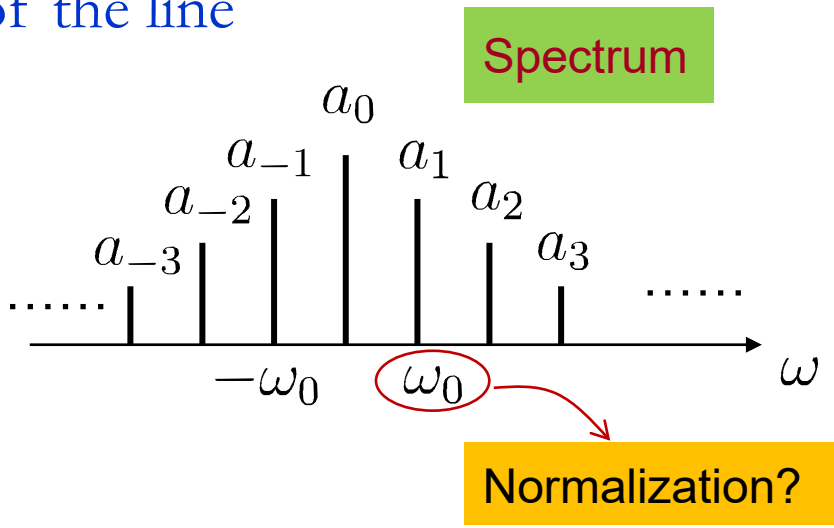
phase



Concept of Spectrum

- It is seen that the harmonic components of FS are only different in coefficient and frequency
  - Use coefficient as the complex amplitude of the line
  - Use frequency as the position of the line

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

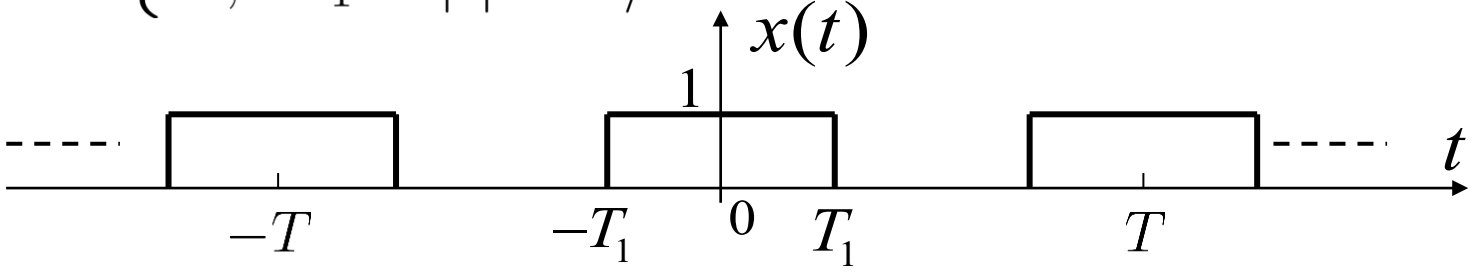


- Spectrum shows the distribution of a_k over the entire frequency and it can completely represent the time domain signal (from frequency domain)

- Example: Periodic square wave

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

A frequently encountered signal in this course!!!



$$a_k = \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega_0 t}dt = \frac{1}{-jk\omega_0 T} e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1}, \quad k \neq 0$$

$$= \frac{2\sin(k\omega_0 T_1)}{Tk\omega_0} = \frac{\sin(k\omega_0 T_1)}{k\pi} = \frac{2T_1}{T}\mathrm{sinc}\left(\frac{2T_1}{T}k\right)$$

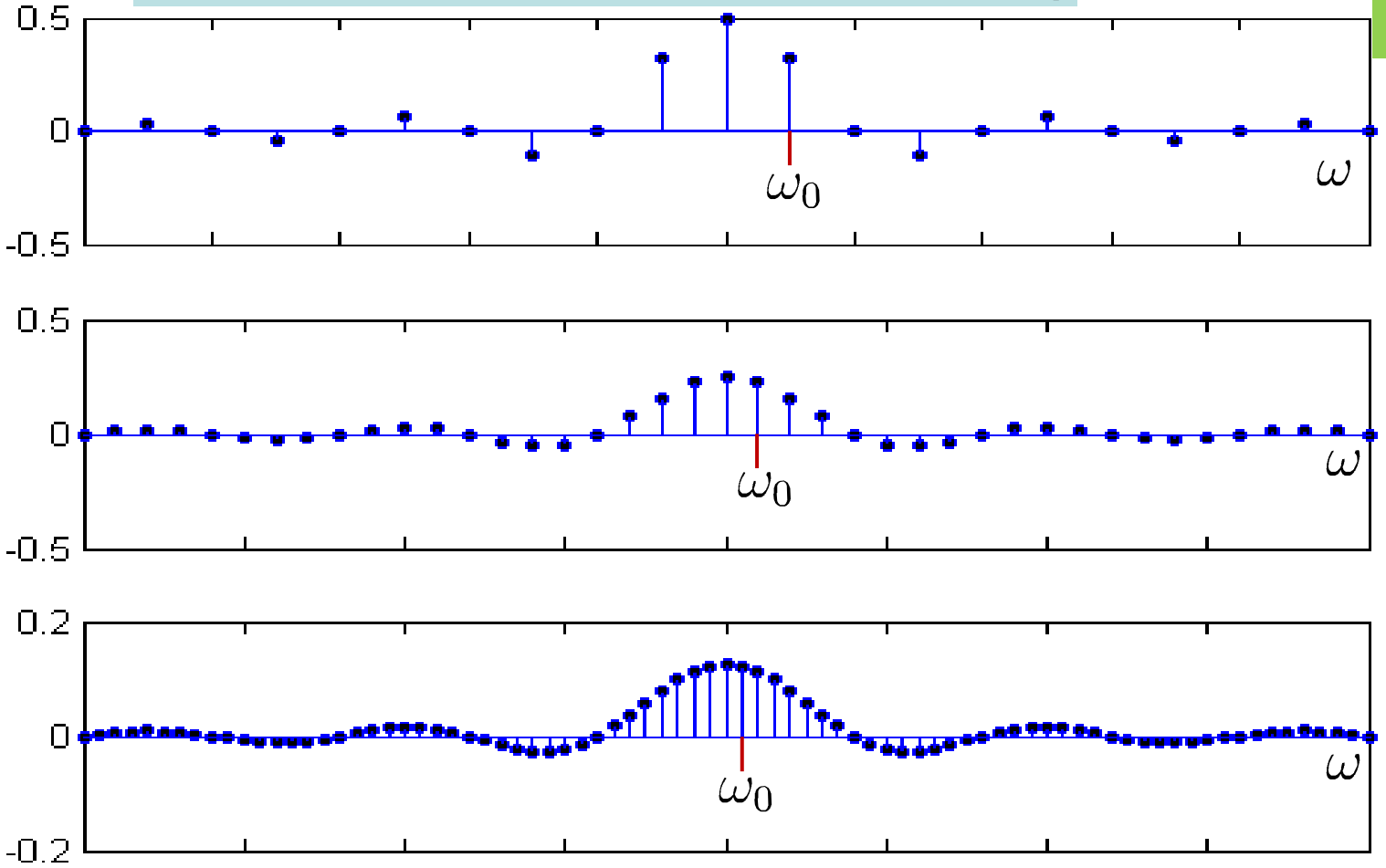
One can draw the spectrum according to a_k

$$\mathrm{sinc}(x) = \frac{\sin \pi x}{\pi x}$$

Fix T_1 while increase T

$\omega_0 = 2\pi/T$

$$a_k = \frac{2\sin(k\omega_0 T_1)}{Tk\omega_0} = \frac{2\sin(\omega T_1)}{\omega T}\bigg|_{\omega=k\omega_0}$$



Peak value

$\frac{2T_1}{T} = \frac{1}{2}$

$\frac{2T_1}{T} = \frac{1}{4}$

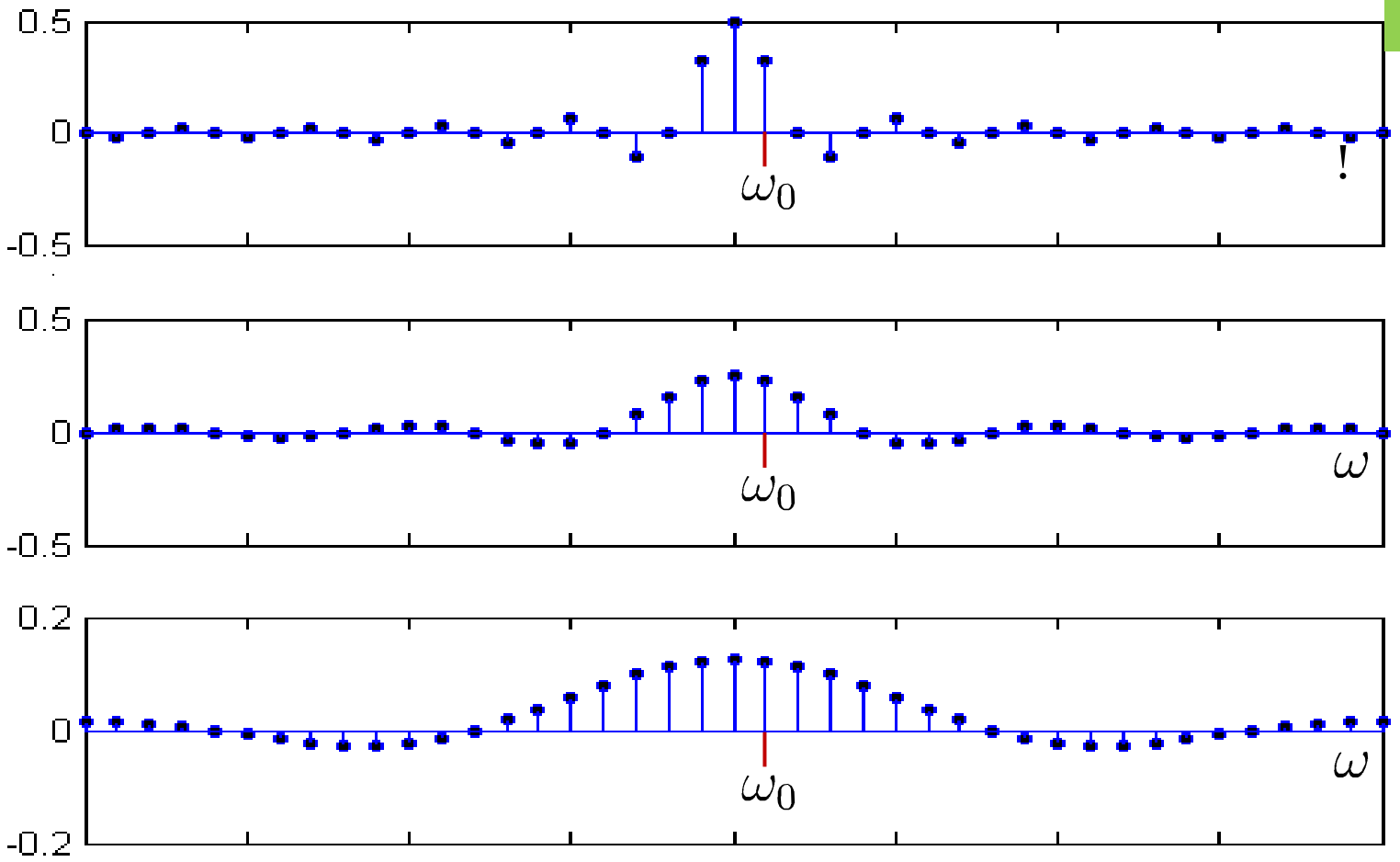
$\frac{2T_1}{T} = \frac{1}{8}$

Fix T while decrease T_1

$\omega_0 = 2\pi/T$

$$a_k = \frac{2\sin(k\omega_0 T_1)}{Tk\omega_0} = \frac{2\sin(\omega T_1)}{\omega T}\bigg|_{\omega=k\omega_0}$$

Peak value



- Features of the spectrum of periodical signals
  - Discrete, even distance $2\pi/T$ between lines (from harmonically related complex exponentials)
  - Fix T_1 but increase T, the distance between lines decreases, the amplitude of lines decrease. However, the wide of mainlobe does not change but the mainlobe contains more harmonics
  - Fix T but decrease T_1, the distance between lines does not change, the amplitude of lines decrease. The wide of mainlobe changes as the mainlobe becomes wider and contains more harmonics

As will be seen later, a periodical signal becomes a non-periodical signal when $T \to +\infty$. Guess what happens to spectrum? (distance, amplitude)

Convergence of Fourier Series

- Fourier's claim: Any periodic signal can be represented by Fourier series.
  - Actually: holds for an extremely large class of periodic signals
  - Surprisingly: Square wave example is discontinuous.
- Define $x_N(t) = \sum_{k=-N}^{N} b_k e^{jk\omega_0 t} \qquad e_N(t) = x(t) - x_N(t)$

  For any integer N!!!
  - Find the best b_k (given integer N) for approximation under minimum mean-square error criterion (MMSE). We obtain

$$E_N = \int_T |e_N(t)|^2 dt$$

$$= \int_T \left(x(t) - \sum_{k=-N}^{k=N} b_k e^{jk\omega_0 t} \right) \left(x(t) - \sum_{k=-N}^{k=N} b_k e^{jk\omega_0 t} \right)^* dt$$

Complex differentiation needed



- Interestingly, it can be proved that $b_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt$
- FS is the best approximation of $x(t)$ under MMSE criterion

- When would a periodic $x(t)$ have FS representation?
  - The value of a_k must exist (be finite)
  - The synthesis equation should converge to $x(t)$,i.e., as $N \to \infty$ $E_N \to 0$.
- Class 1: If the periodic signal has finite energy in one period

$$\int_T |x(t)|^2 dt < \infty$$

Square integrable condition

then a_k exists and $x(t)$ has Fourier series representation that guarantee $E_N \to 0$ as $N \to \infty$

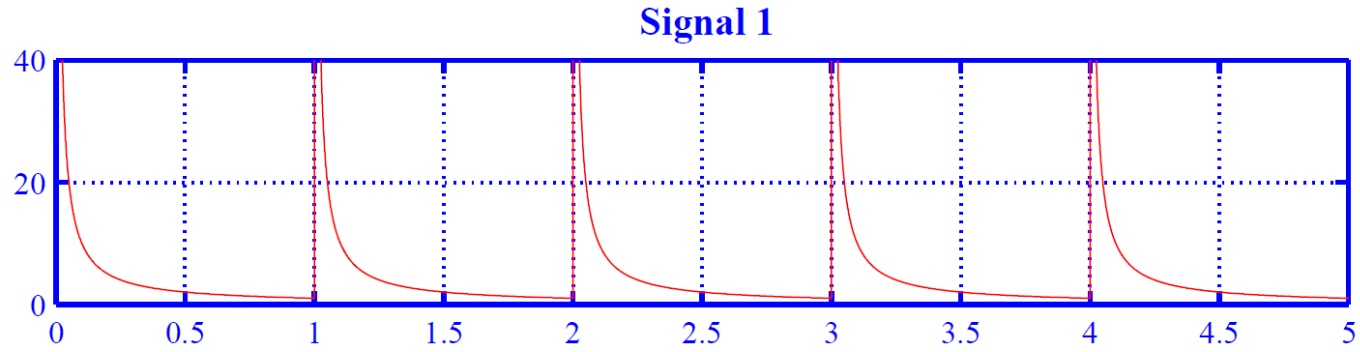
Easy to satisfy in practice

Not exactly equal but no energy difference!!!

- Class 2: Dirchlet conditions
  - Absolutely integrable in one period: $\int_T |x(t)|dt < \infty$

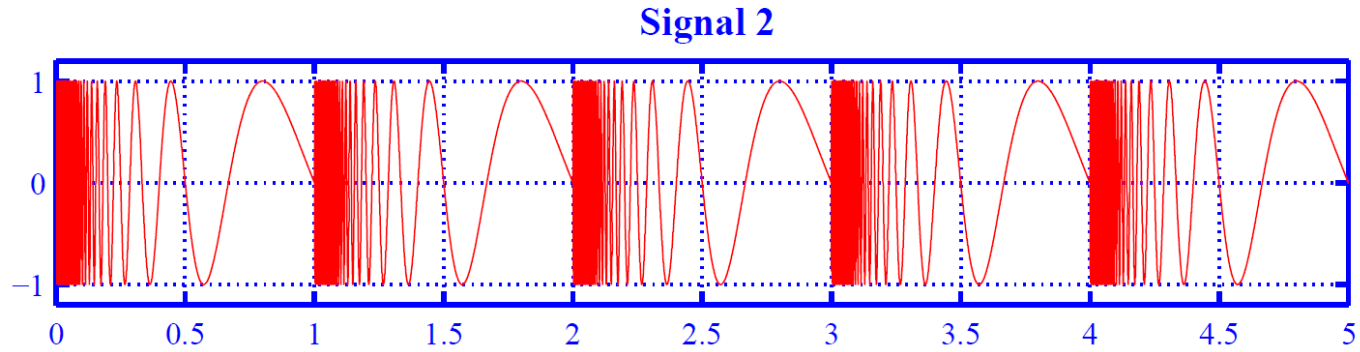
    $\Rightarrow$ $|a_k| \leq \frac{1}{T}\int_T |x(t)e^{-jk\omega_0 t}|dt < \infty$
  - In any finite interval of time, $x(t)$ is of bounded variation, i.e., there are finite number of maxima and minima during any single period.
  - In any finite interval of time, there are only a finite number of discontinuities. Furthermore, each of these discontinuities is finite.

This condition guarantees that $x(t)$ equals to its FS representation, except at isolated values of t for which $x(t)$ is discontinuous



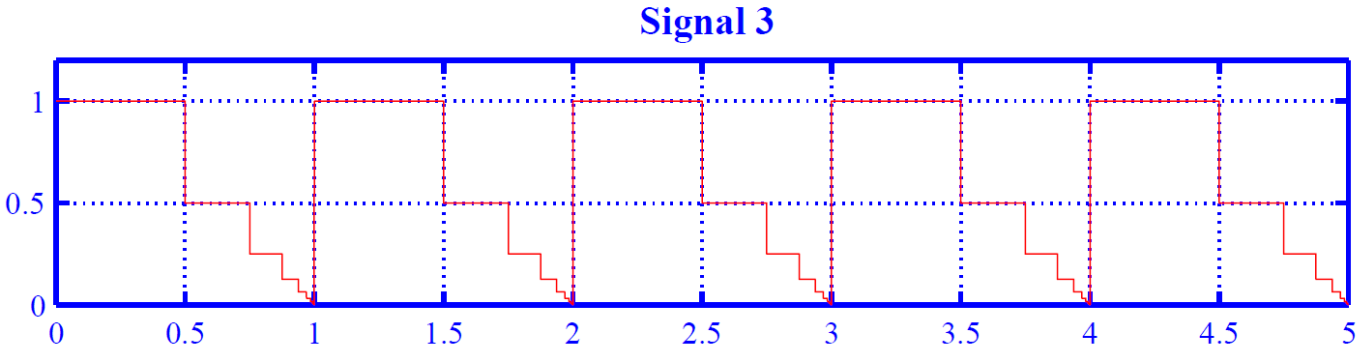
$\frac{1}{t}$

violate condition 1



$\sin\frac{2\pi}{t}$

violate condition 2

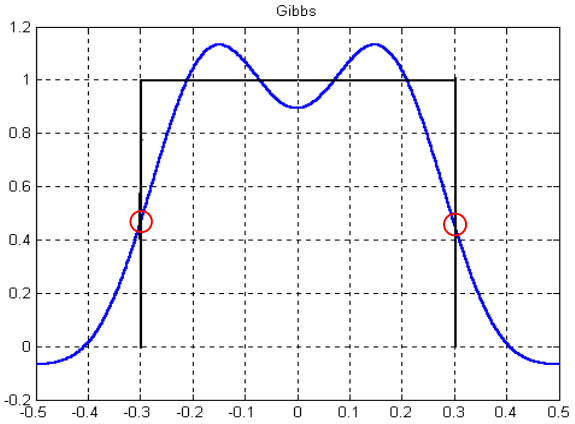


violate condition 3

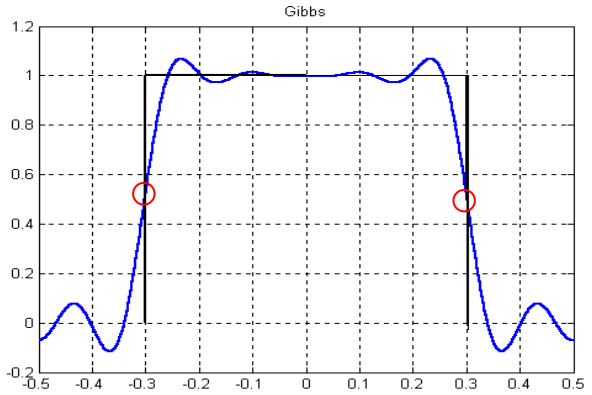
- Some notes
  - Counter-examples are generally pathological
  - For most practical periodical signals we can think the Fourier series exist
  - For a periodic signal that has no discontinuities, the representation converges exactly to every point of $x(t)$
  - For periodic signal with a finite number of discontinuities (in each period), the representation equals $x(t)$ except at isolated points; actually, the series converges to the average value of that the discontinuity-----Difference contain no energy !!! (no effect for LTI system which is characterized by convolution/integration)

Gibbs phenomenon

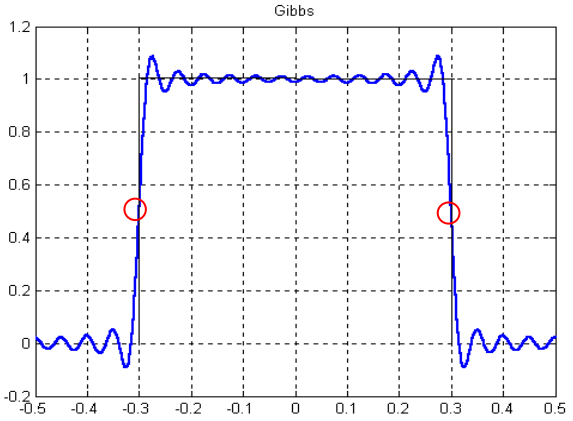
- Under Dirchlet conditions, check how Fourier series converges to a periodic signal with discontinuities.



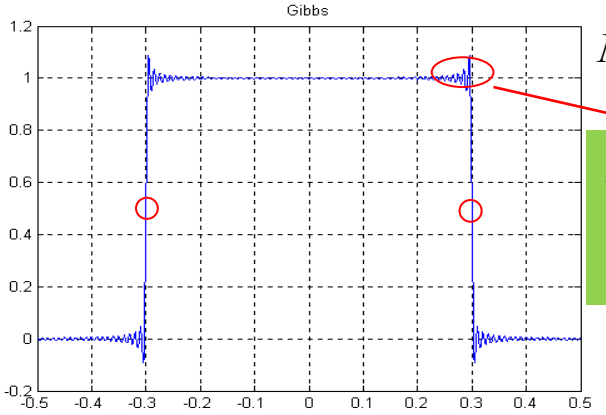
$N = 3$



$N = 7$



$N = 19$



$N = 100$

Around 1.09 regardless of N

- Gibbs phenomenon
  - For any finite N, ripples exist in the vicinity of discontinuity
  - As the number of harmonics increases, the ripples become compressed toward the discontinuity but the peak amplitude of the ripples remains constant (regardless of N).
  - The peak amplitude of the ripples does not decrease as the number of harmonics used in the signal synthesis increases, about 1.09
  - Large enough value of N should be chosen to guarantee the ignorable energy in ripples
  - A direct consequence of using FS to approximate the original signal under MMSE criterion

Properties of Continuous-Time Fourier Series

Assume $x(t)$ and $y(t)$ have the same period T:

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k \qquad y(t) \overset{\mathcal{FS}}{\longleftrightarrow} b_k$$

- Linearity

$$Ax(t) + By(t) \overset{\mathcal{FS}}{\longleftrightarrow} Aa_k + Bb_k$$

- Time shifting

$$x(t - t_0) \overset{\mathcal{FS}}{\longleftrightarrow} a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$$

same amplitude

- Frequency shifting

$$e^{jM\omega_0 t} x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_{k-M}$$

How about $e^{j\omega_s t} x(t)$?

- Time reversal

$$x(-t) \overset{\mathcal{FS}}{\longleftrightarrow} a_{-k}$$

⇒ Even signal $a_k = a_{-k}$ Odd signal $a_k = -a_{-k}$

Properties of Continuous-Time Fourier Series

– Time scaling

$$x(\alpha t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(\alpha\omega_0)t}$$

$$x(\alpha t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

How about complex α ?

Although the FS coefficients have not changed, the FS representation has changed because of the change in the fundamental frequency

– Multiplication

$$x(t)y(t) \overset{\mathcal{FS}}{\longleftrightarrow} \sum_{l=-\infty}^{\infty} a_l b_{k-l} = \{a_k\} * \{b_k\}$$

Same period

– Periodic convolution

$$\int_T x(\tau)y(t-\tau)d\tau \overset{\mathcal{FS}}{\longleftrightarrow} Ta_k b_k$$

– Differentiation

$$\frac{dx(t)}{dt} \overset{\mathcal{FS}}{\longleftrightarrow} jk\omega_0 a_k$$

Check k=0

- Conjugation

$$x^*(t) \overset{\mathcal{FS}}{\longleftrightarrow} a^*_{-k}$$

  - Real signal: $a_{-k} = a_k^*$ Implies many

$$a_k = A_k e^{j\theta_k} = B_k + jC_k$$

  - Real and even signal: $a_k = a_k^*$ Real and Even too!
  - Real and odd signal: $a_k = -a_k^*$ Purely imaginary and odd

- Parseval's relation

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

Average power in one period

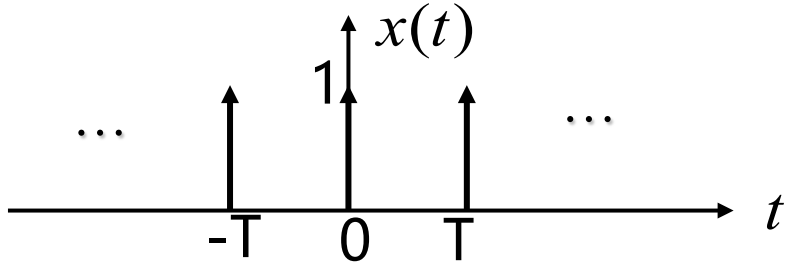
  - $\frac{1}{T}\int_T |a_k e^{jk\omega_0 t}|^2 dt = |a_k|^2$ Average power in the kth harmonic

Some insight one what you can get from spectrum!!!

Examples

- Example: impulse train

$$x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT)$$



  - From definition

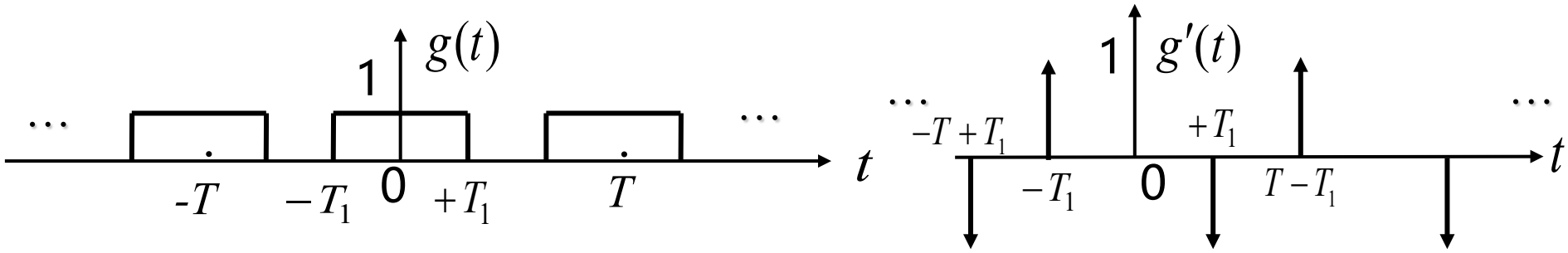
$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)e^{-jk\omega_0 t}dt = \frac{1}{T}$$

  - Hence, another representation of impulse train is

$$x(t) = \sum_{k=-\infty}^{+\infty} \frac{1}{T}e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} \delta(t - kT)$$

Continuous functions can build singular functions!!!

- Example: periodic square wave, again



  - Let $g(t) \overset{FS}{\longleftrightarrow} c_k \quad g'(t) \overset{FS}{\longleftrightarrow} b_k$
  - From differentiation property $b_k = jk\omega_0 c_k$
  - From shifting property $b_k = a_k e^{jk\omega_o T_1} - a_k e^{-jk\omega_o T_1}$
  - From previous example $a_k = 1/T$
  - Hence

a_k from previous example

$$c_k = \frac{b_k}{jk\omega_0} = \frac{2\sin k\omega_o T_1}{k\omega_0 T} = \frac{\sin k\omega_o T_1}{k\pi} = \frac{2T_1}{T}\frac{\sin k\omega_0 T_1}{k\omega_0 T_1}$$

Fourier Series Representation of Discrete-Time Periodic Signals (DTFS)

- Parallel to continuous case, but with difference
  - $x[n] = x[n+N] \quad \omega_0 = 2\pi/N$
  - Harmonic components $\phi_k[n] = e^{jk\omega_0 n}$, $k = 0, \pm 1, \pm 2, \ldots$
  - Build $x[n]$ from N harmonics

Only N distinct harmonics

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$$

where $k =< N >$ represents ANY successive N integers.

  - Clearly, DTFS can build periodic signals
- Determine the FS coefficients a_k
  - Different from continuous case, here we have N linear equations to solve
  - A direct consequence is that there is no convergence problem

$$x[n] = \sum_{k=<N>} a_k e^{jk\frac{2\pi}{N}n}$$

$$\begin{bmatrix} x[1] \\ x[2] \\ \vdots \\ x[N] \end{bmatrix} = \begin{bmatrix} e^{j\frac{2\pi}{N}} & e^{j2\frac{2\pi}{N}} & \cdots & e^{jN\frac{2\pi}{N}} \\ e^{j\frac{2\pi}{N}2} & e^{j2\frac{2\pi}{N}2} & \cdots & e^{jN\frac{2\pi}{N}2} \\ \vdots & \vdots & \ddots & \vdots \\ e^{j\frac{2\pi}{N}N} & e^{j2\frac{2\pi}{N}N} & \cdots & e^{jN\frac{2\pi}{N}N} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \\ a_N \end{bmatrix}$$

How about start with $x[0]$?

First harmonic

Second harmonic

Nth harmonic

– Alternative and simpler approach: Multiplying $e^{-jr\omega_0 n}$ and summing over n:

$$\sum_{n=<N>} x[n]e^{-jr\omega_0 n} = \sum_{n=<N>}\sum_{k=<N>} a_k e^{j(k-r)\omega_0 n}$$

$$= \sum_{k=<N>} a_k \sum_{n=<N>} e^{j(k-r)\omega_0 n} = N a_r$$

$N\delta[k-r]$

where we use the following property

$$\sum_{n=<N>} e^{jp\omega_0 n} = \sum_{n=<N>} e^{jp(2\pi/N)n} = \begin{cases} N, & p = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$$

– Hence $a_k = \frac{1}{N}\sum_{n=<N>} x[n]e^{-jk\omega_0 n}$

- To summarize, if the periodic signal $x[n]$ has a Fourier series representation, then

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk(2\pi/N)n}$$

  - $a_k = a_{k+N}$ is also periodic with period N but only consecutive N of them are used to construct $x[n]$
  - Normally we take $k = 0, 1, \ldots, N-1$

Check real $x[n]$

- Example: $x[n] = \sin \omega_0 n$
  - $\omega_0 = \dfrac{2\pi}{N} \qquad x[n] = \dfrac{e^{j(2\pi/N)n} - e^{-j(2\pi/N)n}}{2j}$

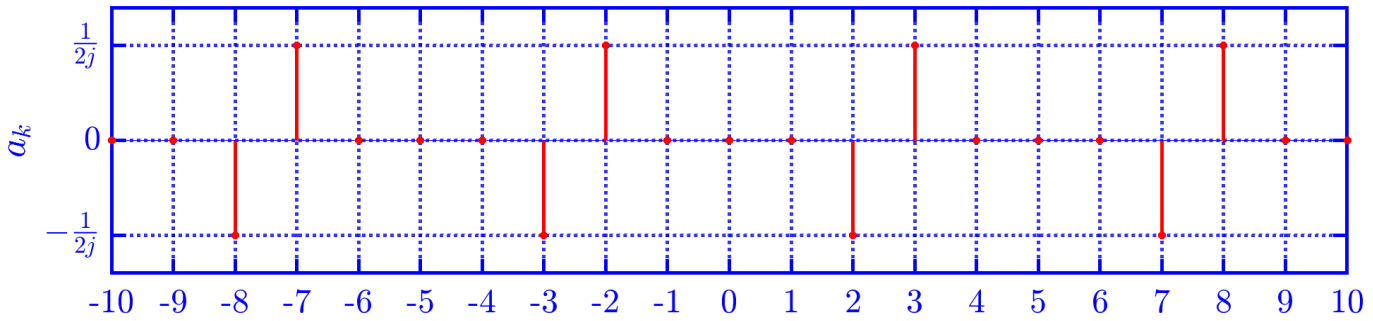
$$\Rightarrow a_1 = \frac{1}{2j} = a_{N+1} \qquad a_{-1} = -\frac{1}{2j} = a_{N-1}$$

  - $\omega_0 = \dfrac{2\pi M}{N} \qquad x[n] = \dfrac{e^{j(2\pi M/N)n} - e^{-j(2\pi M/N)n}}{2j}$

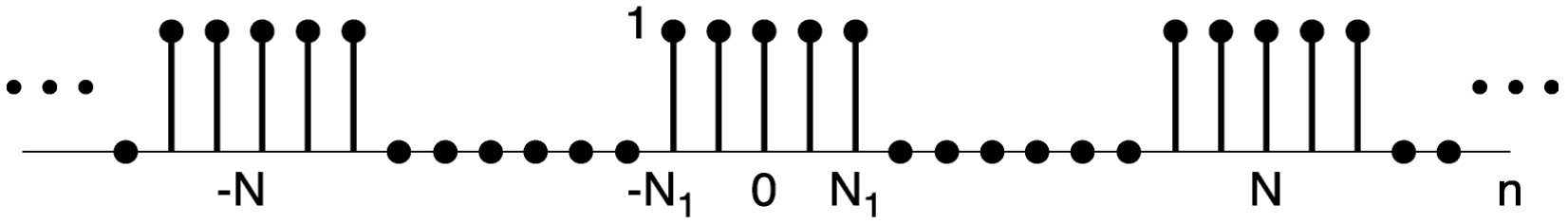
ω_0 cannot take other value！！

$$\Rightarrow a_M = \frac{1}{2j} = a_{N+M} \qquad a_{-M} = -\frac{1}{2j} = a_{N-M}$$

M=3
N=5



- Example: square wave $x[n]$



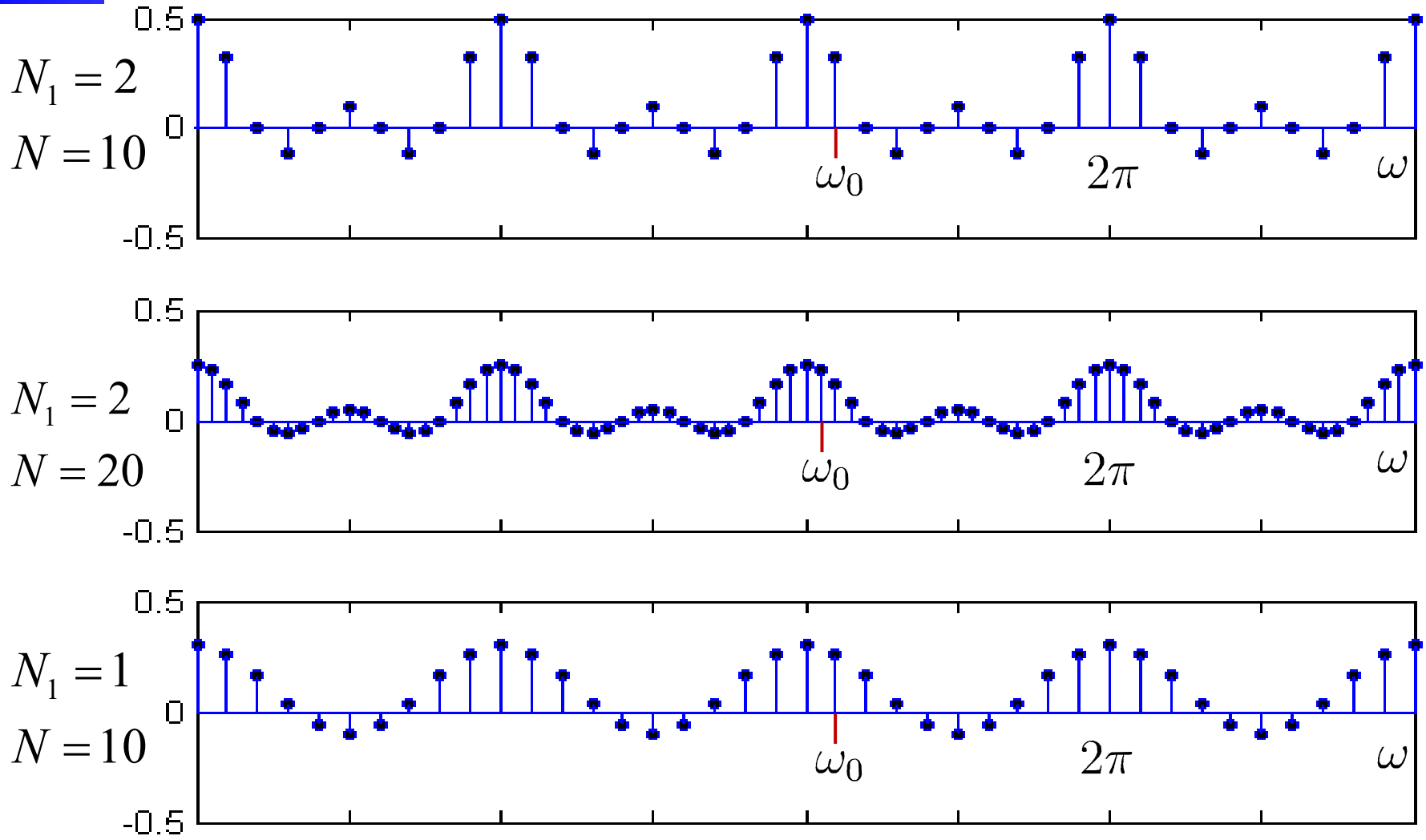
$$a_k = \frac{1}{N}\sum_{n=-N_1}^{N_1} e^{-jk(2\pi/N)n} = \frac{1}{N}\frac{e^{jk\omega_0 N_1} - e^{-jk\omega_0(N_1+1)}}{1-e^{-jk\omega_0}}$$

$$= \frac{1}{N}\frac{\sin[2\pi k(N_1+1/2)/N)]}{\sin(\pi k/N)}, \quad k \neq 0, \pm N, \ldots$$

$$a_k = \frac{1}{N}\sum_{n=-N_1}^{N_1} 1 = \frac{2N_1+1}{N}, \quad k = 0, \pm N, \ldots$$

$$a_k = \frac{1}{N}\frac{\sin[(2N_1+1)\frac{k\omega_0}{2}]}{\sin(\frac{k\omega_0}{2})}$$

The envelope of a_k has the form $\left.\frac{\sin((2N_1+1)\omega/2)}{N\sin(\omega/2)}\right|_{\omega=k\omega_0}$



- Spectrum property of periodic square wave
  - Fix N_1 and increase N, the envelope of the spectrum does not change, but the amplitude will decrease and the spectrum becomes dense
  - Fix N and change N_1, envelop $\frac{\sin((2N_1+1)\omega/2)}{N\sin(\omega/2)}$ will change.

When N_1 decreases, the main lobe will be come wider, ie., $(2N_1+1)\omega = \pm\pi$, which is very similar to the continuous-time periodic square wave. However, the difference is that DTFS is also periodic

Convergence of DTFS

- Since DFTS is the summation of limited number of harmonics, and a_k is also obtained from the summation of limited components, there is no convergence problem
  - There is no Gibbs problem either.
  - The reason behind is transforming N independent values from the time domain to another N discrete values in frequency domain and backwards is feasible.
  - However, for CTFS, since there is infinite values in time domain, we cannot recover these values by finite number of coefficients in frequency domain. Hence, we needs large number of CTFS to better approximate time domain signal and then there is convergence problem.

Properties of Discrete-Time Fourier Series

Notations: $x[n] \overset{\mathcal{FS}}{\longleftrightarrow} a_k \qquad y[n] \overset{\mathcal{FS}}{\longleftrightarrow} b_k$ Same period N

- Most properties are similar to continuous-time case
- Periodic convolution $\sum_{r=<N>} x[r]y[n-r] \overset{\mathcal{FS}}{\longleftrightarrow} Na_k b_k$
- Multiplication $x[n]y[n] \overset{\mathcal{FS}}{\longleftrightarrow} \sum_{l=<N>} a_l b_{k-l}$ Periodic convolution
- First difference $x[n] - x[n-1] \overset{\mathcal{FS}}{\longleftrightarrow} (1 - e^{-jk\omega_0})a_k$
- Parseval's relation

$$\frac{1}{N} \sum_{n=<N>} |x[n]|^2 = \sum_{k=<N>} |a_k|^2$$

See all properties from textbook

- Example: Suppose we are given the following facts:
  - $x[n]$ is periodic with period $N = 6$
  - $\sum_{n=0}^{5} x[n] = 2$
  - $\sum_{n=2}^{7} (-1)^n x[n] = 1$
  - $x[n]$ has the minimum power per period

Find x[n]

From fact 2 we know $a_0 = 2/6 = 1/3$

Fact 3 can be rewritten as

$$\frac{1}{6}\sum_{n=2}^{7} e^{-jn\pi} x[n] = \frac{1}{6}\sum_{n=2}^{7} e^{-j3n(2\pi/6)} x[n] = \frac{1}{6} = a_3$$

Fact 4 requires other coefficient to be zero (from Paseval's relation)

$$x[n] = 1/3 + e^{j3(2\pi/6)n}/6 = 1 + (-1)^n/6$$

Fourier Series and LTI Systems

- LTI system only scales the input complex exponentials
  - $e^{st} \longrightarrow y(t) = H(s)e^{st} \quad H(s) = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$
  - $z^n \longrightarrow y[n] = H(z)z^n \quad H(z) = \sum_{k=-\infty}^{\infty} h[k]z^{-k}$

  } system functions

- Let's consider special case
  - $s = j\omega, \; H(j\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau$
  - $z = e^{j\omega}, \; H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-jk\omega}$

  } frequency response

Periodic with 2π

Viewed as function of frequency ω

Will see very soon ☺

- LTI response for periodic signals
  - $$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

    Also periodical

    New coefficients
  - $$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} \longrightarrow y[n] = \sum_{k=<N>} a_k H(e^{jk\omega_0}) e^{jk\omega_0 n}$$
  - The effect of LTI system is purely to modify the Fourier series coefficients by multiplying the value of the frequency response at the corresponding frequency $k\omega_0$ with different weight, i.e., modify the lines in spectrum with different weight
  - Hence, the output of LTI system is also periodic

- Example: For LTI system $h[n] = \alpha^n u[n],\ -1 < \alpha < 1$, compute the response to $x[n] = \cos(\frac{2\pi}{N} n)$
  - $x[n] = \frac{1}{2}[e^{j\frac{2\pi}{N}n} + e^{-j\frac{2\pi}{N}n}]$,i.e,. $a_1 = a_{-1} = \frac{1}{2}$
  - $H(e^{jk\omega_0}) = \sum_{n=0}^{\infty} \alpha^k e^{-jk\omega_0 n} = \frac{1}{1 - \alpha e^{-jk\omega_0}}$

Stable systems

$$H(e^{j\frac{2\pi}{N}}) = \frac{1}{1 - \alpha e^{-j\frac{2\pi}{N}}} \qquad H(e^{-j\frac{2\pi}{N}}) = \frac{1}{1 - \alpha e^{j\frac{2\pi}{N}}}$$

$$b_1 = \frac{1/2}{1 - \alpha e^{-j\frac{2\pi}{N}}} \qquad b_{-1} = \frac{1/2}{1 - \alpha e^{j\frac{2\pi}{N}}} \qquad y[n] = \sum_{k=\pm 1} b_k e^{j\frac{2\pi}{N}kn}$$

See textbook

Summary

- Complex exponentials are the eigen-functions of any LTI systems
- Represent periodic signals by FS and achieve the frequency domain decomposition of signals
- Analyze the property of periodic square wave as well as its change with parameters
- Discuss over the property of FS and find out the similarity between CTFS and DTFS
- Present the response of LTI system to the input periodic signals.

Homework #3

- 3.8, 3.14, 3.23, 3.25, 3.28, 3.34, 3.36, 3.44, 3.47, 3.50
- Try whenever you think you can handle.
- Try finish all questions after the chapter.
- Homework is always due 1 week after each Chapter is finished.